Class 18 Learning Activity

Liquid-phase reaction (1) is exothermic with a heat of reaction equal to –16.6 kcal mol^{-1}. If a solution containing only A at a concentration of 2 M and a temperature of 20 °C reacts adiabatically in a BSTR, the temperature increases to 110 °C. The reaction is first order in the concentration of A as shown in equation (2). The pre-exponential factor is 2.5 x 10^8 L mol^{-1} min^{-1}, and the activation energy is 14.3 kcal mol^{-1}.

$$A \rightarrow Z \tag{1}$$

$$r_1 = k_1 C_A \tag{2}$$

Adiabatic operation has undesirable side effects, making isothermal operation preferable. To do so, using a 0.4 L BSTR with a 0.1 L shell has been proposed. The heat transfer area is 32.5 cm^2, and the heat transfer coefficient is 0.2 cal min^{-1} cm^{-2} K^{-1}. Chilled water at 10 °C would initially fill the shell and be fed to it. The water can be taken to have a density of 1 g cm^{-3} and a heat capacity of 1 cal g^{-1} K^{-1}. The reactor temperature will not be close to isothermal if a fixed coolant flow rate is used. Consequently it has been proposed to set the cooling water flow rate to 217 g min^{-1} initially and to linearly decrease it by 212 g min^{-1} over the duration of the 60 min reaction time.

Assuming the reacting fluid density to equal 879 g L^{-1} and its heat capacity to equal 0.42 cal g^{-1} K^{-1}, plot the conversion, reacting fluid temperature, and cooling water temperature over the course of the reaction and comment on how closely the operation approaches being isothermal.